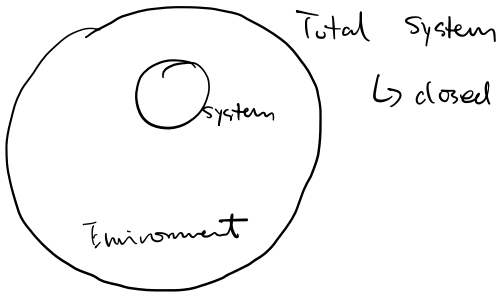


整个 total system 的 dynamics 太复杂了，希望找到 reduced 的 system 的 dynamics 方程，which is an open quantum system.

Density matrix

$$\longrightarrow \begin{cases} \mathrm{Tr}[\rho] = 1 \\ \rho > 0 \quad \text{, i.e. } \forall |\psi\rangle : \langle\psi|\rho|\psi\rangle > 0 \end{cases}$$

$$\rho = \sum_i p_i |\psi_i\rangle\langle\psi_i|$$

$$\Longrightarrow \rho(t) = \sum_i p_i U_t |\psi_i\rangle\langle\psi_i| U_t^\dagger = U_t \rho U_t^\dagger = e^{-iHt} \rho e^{iHt}$$

$$\frac{d}{dt}|\psi(t)\rangle = -iH|\psi(t)\rangle$$

$$\Longrightarrow \frac{d}{dt}\rho(t) = \frac{d}{dt}\left(e^{-iHt}\rho e^{iHt}\right)$$

$$= -iH e^{-iHt}\rho e^{iHt} + e^{-iHt}\rho (iH) e^{iHt}$$

$$= i\left(\rho(t)H - H\rho(t)\right) = -i[H, \rho(t)]$$

i.e. $\dot{\rho}(t) = -i[H, \rho(t)] \equiv \mathcal{L}\rho$ von Neumann equation

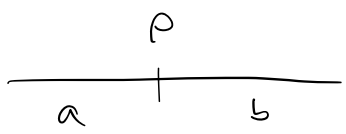
↳ Liouvillian superoperator

$$\frac{d}{dt}\left(\mathrm{Tr}[\rho^2]\right) = \mathrm{Tr}\left(\frac{d\rho^2}{dt}\right) = \mathrm{Tr}(2\rho\dot{\rho}) = -2i\,\mathrm{Tr}[\rho(H\rho - \rho H)] = 0$$

↳ purity

时间演化不改变量子系统的 purity.

Reduced density matrix



$$\rho_a = \mathrm{Tr}_b[\rho]$$

$$= \mathrm{Tr}_b\left[\sum_{i,j,k,l} |a_i\rangle\langle a_j| \otimes |b_k\rangle\langle b_l|\right]$$

$$\equiv \sum_{i,j} |a_i\rangle\langle a_j| \,\mathrm{Tr}\left[\sum_{k,l} |b_k\rangle\langle b_l|\right]$$

e.g. two-level system

$$\mathcal{H} = \mathcal{H}_2 \otimes \mathcal{H}_2 \quad \text{with a basis:}$$

$$\{ |00\rangle = |0\rangle_1 \otimes |0\rangle_2,\ |01\rangle = |0\rangle_1 \otimes |1\rangle_2,\ |10\rangle = |1\rangle_1 \otimes |0\rangle_2,\ |11\rangle = |1\rangle_1 \otimes |1\rangle_2 \}$$

$|\psi\rangle_G = |00\rangle$ both ground state, seperable state

$$|\psi\rangle_S = \frac{1}{\sqrt{2}}(|0\rangle_1 + |1\rangle_1) \otimes \frac{1}{\sqrt{2}}(|0\rangle_2 + |1\rangle_2)$$

$$= \frac{1}{2}(|00\rangle + |10\rangle + |01\rangle + |11\rangle) \quad \text{seperable state}$$

$$|\psi\rangle_E = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) \quad \text{entangled state}$$

$$\rho_E = |\psi\rangle_E \langle\psi| = \frac{1}{2}(|00\rangle\langle 00| + |00\rangle\langle 11| + |11\rangle\langle 00| + |11\rangle\langle 11|)$$

$$\rho_E^{(1)} = {}_2\langle 0|\rho_E|0\rangle_2 + {}_2\langle 1|\rho_E|1\rangle_2 = \frac{1}{2}(|00\rangle\langle 00| + |11\rangle\langle 11|)$$

The Fock-Liouville Hilbert Space, The Liouville Superoperator

Some linear combinations of density matrices are valid density matrices (as long as they preserve positivity and trace 1).

$\Longrightarrow$ 可以将这一个 density matrice 与 Hilbert space by 给出 scalar product 的定义. 这个空间中的 vector 是矩阵 ($\rho \longrightarrow |\rho\rangle\rangle$). Called Fock-Liouville Space (FLS)

$$\langle\langle \phi | \rho \rangle\rangle \equiv \mathrm{Tr}[\phi^\dagger \rho]$$

e.g. time evolution of a two-level atom

$$H = E|1\rangle\langle 1| + \Omega(|0\rangle\langle 1| + |1\rangle\langle 0|)$$

$$|\rho\rangle\rangle = \begin{pmatrix} \rho_{00} \\ \rho_{01} \\ \rho_{10} \\ \rho_{11} \end{pmatrix} \qquad \tilde{\mathcal{L}} = \begin{pmatrix} 0 & i\Omega & -i\Omega & 0 \\ i\Omega & iE & 0 & -i\Omega \\ -i\Omega & 0 & -iE & i\Omega \\ 0 & -i\Omega & i\Omega & 0 \end{pmatrix}$$

$$\Longrightarrow \begin{pmatrix} \dot{\rho}_{00} \\ \dot{\rho}_{01} \\ \dot{\rho}_{10} \\ \dot{\rho}_{11} \end{pmatrix} = \begin{pmatrix} 0 & i\Omega & -i\Omega & 0 \\ i\Omega & i\Sigma & 0 & -i\Omega \\ -i\Omega & 0 & -i\Sigma & i\Omega \\ 0 & -i\Omega & i\Omega & 0 \end{pmatrix} \begin{pmatrix} \rho_{00} \\ \rho_{01} \\ \rho_{10} \\ \rho_{11} \end{pmatrix}$$